

Machine Failure Prediction Model Report

1. Abstract

This report presents the development and evaluation of a binary classifier for predicting industrial machine failures from multi-sensor data. Using a 944-row dataset with nine sensor features, three models were evaluated — a majority-class baseline, Logistic Regression, and Random Forest. The Random Forest classifier achieved the best performance (accuracy 93.1%, ROC-AUC 0.975), and feature importance analysis identified volatile organic compounds (VOC), air quality (AQ), and ultrasonic proximity (USS) as the strongest predictors of failure. The results demonstrate that sensor fusion is a viable approach for early failure detection.

2. Problem Statement

Unexpected machine failures cause costly downtime, safety hazards, and disrupted production. This project explores whether multi-sensor readings — temperature, air quality, vibration-related signals, and other indicators — can predict machine failure before it occurs, enabling proactive maintenance.

3. Dataset

The dataset contains 944 rows and 10 columns (9 sensor features + 1 binary target). Each row is an independent observation of a machine's operating state.

Feature	Description
footfall	Number of people/objects passing the machine
tempMode	Temperature mode setting of the machine
AQ	Air quality index near the machine
USS	Ultrasonic sensor proximity measurement
CS	Current sensor reading (electrical current)
VOC	Volatile organic compounds level
RP	Rotational position / RPM
IP	Input pressure to the machine
Temperature	Operating temperature of the machine

fail	Target — 1 = failure, 0 = no failure
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Class balance: ~42% failures, ~58% non-failures.

Source: umerrtx. (2023). *Machine Failure Prediction Using Sensor Data*. Kaggle.

<https://www.kaggle.com/datasets/umerrtx/machine-failure-prediction-using-sensor-data>

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Limitation: This dataset is *tabular* — each row is an independent snapshot. It does not contain time-series information and therefore cannot model gradual degradation over time. For temporal predictive maintenance, the AI4I 2020 dataset is the standard benchmark.

4. Methodology

4.1 Preprocessing

- The dataset contained **no missing values** (verified via `isnull().sum()`).
- **Standard scaling** was applied to all features using `StandardScaler` to ensure features with different ranges (e.g., `footfall`: 0–7300 vs. `tempMode`: 0–7) contribute equally to model training.
- A **stratified 80/20 train/test split** preserved the class distribution: 755 training samples, 189 test samples.

4.2 Models Evaluated

1. **Majority-class baseline** (`DummyClassifier`) — reference point for comparison.
2. **Logistic Regression** (`max_iter=1000`) — linear baseline.
3. **Random Forest** (`n_estimators=100`) — ensemble model capturing non-linear interactions.

4.3 Evaluation Metrics

- Accuracy, Precision, Recall, F1-score
- ROC-AUC (Area Under the ROC Curve)
- Confusion matrix
- 5-fold cross-validation for stability assessment

Note: Regression metrics such as R^2 and Mean Absolute Error are **not applicable** to this binary classification task and were excluded from evaluation.

5. Results

5.1 Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Majority Baseline	0.583	0.000	0.000	0.000	0.500
Logistic Regression	0.9365	0.9351	0.9114	0.9231	0.9780
Random Forest	0.9312	0.9231	0.9114	0.9172	0.9753

Both machine learning models substantially outperform the majority-class baseline. Random Forest was selected as the final model based on its high F1 and ROC-AUC, and its ability to capture non-linear feature interactions.

5.2 Confusion Matrix (Random Forest)

	Predicted: No Failure	Predicted: Failure
Actual: No Failure	104 (TN)	6 (FP)
Actual: Failure	7 (FN)	72 (TP)

On the test set, the model missed 7 failures (false negatives) and raised 6 false alarms (false positives).

5.3 Cross-Validation (5-fold F1)

Model	Mean F1	Std Dev
Logistic Regression	0.8788	± 0.0247
Random Forest	0.8714	± 0.0236

The two models perform comparably under cross-validation, with Logistic Regression achieving a slightly higher mean F1.

5.4 Feature Importance (Random Forest)

The top predictors of machine failure were:

Feature	Importance
VOC	~0.45
AQ	~0.15

USS	~0.13
CS	~0.06
RP	~0.06
footfall	~0.05
Temperature	~0.04
IP	~0.02
tempMode	~0.02

Interpretation: VOC dominates — chemically degrading machinery releases volatile compounds before mechanical failure. AQ and USS follow, indicating that environmental and proximity signals are also informative. Interestingly, AQ showed a strong linear correlation with failure (Pearson $r = 0.58$) but only modest Random Forest importance — the tree-based model captured non-linear interactions that correlation alone cannot detect.

6. Discussion

The project demonstrates three key findings:

1. **Sensor fusion works.** Combining nine heterogeneous signals yields 93% accuracy and ROC-AUC 0.975 for binary failure prediction.
2. **Non-linear models matter.** Random Forest captured feature interactions (e.g., the mismatch between AQ's high correlation but modest importance) that Logistic Regression could not.
3. **Feature importance has physical meaning.** VOC being the top predictor aligns with industrial intuition: chemical degradation precedes mechanical failure.

Limitations:

- The dataset is tabular, not temporal — the model cannot predict *future* failures from degradation trends.
- The dataset is small (944 rows) and may not represent the full diversity of industrial conditions.
- The class balance (42% failures) is unusually high for real-world predictive maintenance, where failures are typically <5%.

7. Conclusion and Future Work

This project built and evaluated a machine failure prediction model using supervised classification. By combining preprocessing, multiple model comparison, and rigorous evaluation (ROC-AUC, cross-validation, feature importance), the project establishes a reproducible baseline for sensor-based failure prediction.

Future directions:

1. **Benchmark on AI4I 2020** — a 10,000-row standard dataset with ~3.4% failure rate, testing the approach under severe class imbalance.
2. **Time-series modeling** — apply LSTM or Temporal CNN architectures to capture degradation over time.
3. **Interpretability** — add SHAP values for per-prediction feature attribution.
4. **Threshold tuning** — optimize the precision-recall trade-off for safety-critical deployment, where false negatives are costlier than false positives.
5. **Uncertainty quantification** — use Bayesian methods to flag low-confidence predictions.

8. References

1. umerrtx. (2023). *Machine Failure Prediction Using Sensor Data*. Kaggle.
<https://www.kaggle.com/datasets/umerrtx/machine-failure-prediction-using-sensor-data>
2. Matzka, S. (2020). *AI4I 2020 Predictive Maintenance Dataset*. UCI Machine Learning Repository.
<https://archive.ics.uci.edu/ml/datasets/AI4I+2020+Predictive+Maintenance+Dataset>

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